

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION)

CLASS: B.TECH.
BRANCH: MECH., CIVIL, CHEM., PIE, BIOTECH

SEMESTER : III/ADD
SESSION : MO/2023

SUBJECT: MA203 NUMERICAL METHODS

TIME: 3 Hours

FULL MARKS: 50

INSTRUCTIONS:

1. The question paper contains 5 questions each of 10 marks and total 50 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

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|--|----|----|
| Q.1(a) Execute five iterations of Bisection method to find the positive real root of $f(x) = x + \ln x - 2 = 0$, starting with $a = 1$ and $b = 2$ as the initial approximations. [5] | 1 | 3 |
| Q.1(b) Taking $x_0 = 3$ as the initial guess, determine the positive real root of $f(x) = x^3 - 8x - 4$ correct to three decimal places using Newton Raphson method. [5] | 1 | 3 |
| Q.2(a) Identify the values of x_1, x_2, x_3 from the following system of linear equations with the use of Gauss Elimination method:
$5x_1 - x_2 = 9; -x_1 + 5x_2 - x_3 = 4; -x_2 + 5x_3 = -6$ [5] | 2 | 3 |
| Q.2(b) Compute two iterations each of Gauss-Jacobi and Gauss-Seidel methods for the following system, starting with $(x, y)^T = (0, 0)^T$ [5] | 2 | 3 |
| $\begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ | | |
| Q.3(a) Obtain the unique polynomial $P(x)$ of degree two using Lagrange method of interpolation, such that $P(1) = 1, P(3) = 27, P(4) = 64$. [5] | 3 | 3 |
| Q.3(b) Taking reference of the data given below, compute $f(x)$ at $x = 2$ using Newton divided-difference interpolation formula: [5] | 3 | 3 |

$x :$	0	1	4	5
$f(x) :$	8	11	68	123

- Q.4 The velocities (v) of a car in km/min running on a road at time intervals (t) of 5 minutes are given below: [5+5=10] 4 2,3

$t :$	0	5	10	15	20
$v :$	43.1	47.7	52.1	56.4	60.8

- a) Compute the acceleration $a = \frac{dv}{dt}$ of the car at $t = 5$ minute with the use of Newton - Forward Interpolation formula.
- b) Obtain the distance $s = \int_0^{20} v dt$ covered by the car in 20 minutes using Trapezoidal rule and Simpson's One-Third rule.

PTO

- Q.5(a) Solve the differential equation $\frac{dy}{dx} = x^2 + y$, with $y(0) = 1$ at $x = 0.4$ in steps of $h = 0.1$ by Euler's method. Compare the obtained numerical result at $x = 0.4$ with those obtained from the exact solution $y(x) = 3e^x - x^2 - 2x - 2$ by evaluating percentage relative error. [5] 5 2,3
- Q.5(b) State fourth order Runge-Kutta formula to find the solution $y(x_1)$ of the initial value problem $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$, where $x_1 = x_0 + h$ and h is the step size. Hence, apply it to compute $y(0.2)$ correct to four decimal places with $h = 0.2$ from $\frac{dy}{dx} = -xy$, with $y(0) = 1$. [5] 5 1,3

::::::21/11/2023::::::E